

Project Work Brief

Two applications built end-to-end: the PinithUB platform, and an internal Employee Work Tracker

PREPARED BY

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PROJECTS COVERED

PinithUB Platform,
Employee Work Tracker

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PROJECT 1

PinithUB Platform

Unified platform for image & document protection, forensic verification, marketplace, and platform administration

1.1 Overview

PinithUB is a forensic-grade platform for protecting images and documents, verifying their authenticity, and giving creators a marketplace to license their work. Four major pieces of the system have been built out: the public marketing site, an internal CMS to run that site, the pixel-level protection engine that is the core of the product, and a Master Admin console that gives the platform team full operational visibility across users, assets, investigations, and revenue.

1.2 Landing Page (PINITHUB Marketing Site)

A complete public marketing site for the platform, redesigned around an enterprise SaaS narrative with clear calls-to-action into the live product. Rather than a static brochure page, it opens with a product story that walks a visitor through what PINITHUB solves before asking for anything, then funnels them toward two outcomes: watch the embedded product demo, or move into the live PinIT Hub product itself. The brand hierarchy is explicit throughout — PINITHUB as the parent platform, PinIT Hub as the live product, PinIT Vault as a module inside it, and PinIT Exchange/Career flagged as "coming soon" — so the messaging doesn't overpromise features that aren't live yet.

- Built on Next.js 15 (App Router), Tailwind CSS v4, Framer Motion for animation, and three.js for 3D visuals
- Enterprise-narrative redesign: positions PINITHUB as the parent brand and PinIT Hub as the live product
- Product demo video embedded on the hero section with a dedicated "Watch demo" flow
- Split into two independent Vercel deployments — a public surface and an admin surface — so the CMS login is never reachable from the public marketing URL

1.3 Admin / CMS for the Landing Page

A dedicated internal admin app for editing the marketing site's content without a code deploy, isolated from the public site for security.

- Authenticated admin login, separate from the public site's routing entirely (returns 404 on the public deploy)
- Content management for page sections and structured collections
- Inbox for demo requests and newsletter sign-ups captured from the public site
- Admin user management and an audit log of admin actions

- Shares one Postgres database (Prisma) with the public site; content edits go live via on-demand revalidation

1.4 Pixel-Level Protection — Images & Documents

The core forensic engine, and the product's main differentiator. Most "tamper detection" tools only answer yes/no — has this file changed since it was protected. This engine goes further: instead of fingerprinting a file as one single unit, it breaks the image or document into a grid of small regions and cryptographically authenticates each one independently. If a face is swapped, a signature is edited, or a single clause is altered on a scanned page, the system doesn't just flag "this file was tampered with" — it can point to the exact region that changed. That precision is what turns a forensic report into something an investigator or a court can actually act on, rather than a plain pass/fail flag.

- Hierarchical spatial authentication — files are broken into a grid of blocks, each independently keyed and authenticated, so a tampered region can be pinpointed rather than just flagging "the file changed"
- Pixel-level keyed authentication scheme (HKCA) already iterated through two hardened versions (v1 → v1.1, with a stronger binary crypto serialization) without breaking older protected files
- Fragment / splice detection — separately catches a small piece of a protected original (e.g. a cropped face) pasted into an unrelated image, which whole-image comparison would miss
- Document protection extended to PDFs: pages are rasterized and run through the same pixel-level protection pipeline as standalone images, with configurable render quality and a page cap for cost control
- Full investigation tooling on top: tamper localization, forensic scanning, and a unified investigation service that ties matches back to a case an investigator can own and close

1.5 Codebase Consolidation & Merge Resolution

Alongside feature work, a significant piece of engineering went into keeping one coherent codebase as it grew across contributors and environments.

- Consolidated a personal development repository into the company's shared main branch, reconciling two divergent histories into one
- Resolved conflicting changes to the Prisma database schema during branch syncs — the highest-risk conflict point, since a bad resolution there can silently corrupt production data
- Resolved conflicts in shared project documentation (README) after parallel edits on different branches
- Synced local work with the remote branch after histories diverged, using stash and cherry-pick workflows to preserve in-flight changes from both sides rather than discarding either

1.6 Master Admin Dashboard — Whole-Platform Operations

A standalone admin console (its own app, independent of the main product) giving the platform team a single place to operate every part of PinitHUB.

- **Command Center** — live KPIs (users, organizations, assets, DNA-protected files, revenue), activity trends, and critical alerts on one screen
- **Users & Organizations** — full directory with drill-down into any user or org
- **Assets & DNA** — every protected file and its fingerprint history
- **Intelligence (Sentinel)** — investigations, tamper findings, and forensic detections, each resolvable into an owned case
- **Marketplace, Billing & Credits** — subscriptions, revenue, and usage against plan limits
- **Network Intelligence & Web Monitoring** — organizational reach and crawler-based leak detection
- **Identity & Verification, Support & Disputes, Audit Logs, Reports** — the remaining day-to-day operational surfaces, all in one console
- Backed by capability-scoped role-based access control and an admin audit trail, so every admin action is permissioned and logged
- Rebuilt this week as its own dedicated app with a full visual pass — corrected a layout bug that was leaving half the screen blank, and reworked the type scale for legibility on the ops team's displays

1.7 Status Snapshot

Area	Status
Landing Page (public marketing site)	Live
Landing Page Admin / CMS	Live
Pixel-Level Protection — Images	Live
Pixel-Level Protection — Documents (PDF)	Live
Codebase Consolidation (personal → company main)	Complete
Master Admin Dashboard	In Progress — core built, visual pass underway

1.8 Technology Stack

Layer	Technologies
Frontend	React, TypeScript, Next.js 15, Vite, Tailwind CSS, Framer Motion, three.js
Backend	Node.js, Express, TypeScript, FastAPI (Python AI sidecar)
Database	PostgreSQL (Supabase), Prisma ORM
Storage	Supabase Storage (encrypted vault)
Security	Keyed pixel-level authentication (HKCA), AES-256-GCM encryption, JWT auth, RBAC
Deployment	Vercel (frontend/CMS), Render (backend + AI service)

PROJECT 2

Employee Work Tracker

Internal tool for attendance, task assignment, and weekly reporting in one authenticated app

2.1 Overview

A self-contained line-of-business application: a FastAPI + PostgreSQL backend paired with a React/TypeScript frontend, covering role-based access, live dashboards, project and task management, attendance, and structured reporting for a small team.

2.2 Access, Dashboards & Reporting

- JWT-based auth with bcrypt-hashed passwords and two roles — Admin and Employee — enforced on both the API and the frontend's protected/admin-only routes
- Admin dashboard: live headcount, today's attendance split, active project count, pending vs. completed tasks, and weekly-doc submission rate, computed directly from the database
- Six-day attendance trend and per-project completion bars driven by live task counts rather than manually entered progress
- Employee dashboard: a scoped personal view — today's attendance, assigned/completed/pending tasks, and this week's status

2.3 Projects, Tasks & Attendance

- Projects carry a status (active / completed / on hold); tasks carry priority and move through a five-stage pipeline (pending → in progress → blocked / testing → completed)
- Tasks record who they're assigned to, who assigned them, due dates, and completion dates — completing a task automatically moves its project's progress bar
- Attendance is logged per employee per day with check-in/check-out, computed working hours, and a status (present / half-day / absent / leave), with one record per person per day enforced at the database level

2.4 Daily Status, Weekly Docs & Audit Trail

- Daily status log — a standup-style entry per employee per day (today's work, hours logged, blockers, tomorrow's plan), optionally linked to a task
- Weekly documentation — a structured end-of-week report (summary, challenges, solutions, testing done, next week's plan), one per employee per ISO week
- Every material action writes to an activity log surfaced on the admin dashboard's recent-activity feed
- A smoke-test suite exercises the core auth, user, and task flows against a disposable database

2.5 Status Snapshot

Area	Status
Auth & Role-Based Access	Built · Local
Admin & Employee Dashboards	Built · Local
Projects & Tasks	Built · Local

Attendance Tracking	Built · Local
Daily Status & Weekly Docs	Built · Local
Production Deployment	Not Started

2.6 Technology Stack

Layer	Technologies
Frontend	React 19, TypeScript, Vite, Tailwind CSS, Radix UI, TanStack Query, Recharts
Backend	FastAPI (Python), SQLAlchemy 2.0, Alembic
Database	PostgreSQL, SQLAlchemy ORM
Storage	Not applicable — no file storage layer
Security	JWT auth (PyJWT), bcrypt password hashing, role-scoped routes
Deployment	Not yet deployed — local development only

3. Week-by-Week Summary

Week	Focus
Week 1	Landing Page — PINITHUB marketing site build and enterprise-narrative redesign, plus the Landing Page Admin/CMS
Week 2	Employee Work Tracker — auth & roles, admin/employee dashboards, projects & tasks, attendance tracking
Week 3	Pixel-Level Protection — hierarchical spatial authentication, keyed pixel authentication (HKCA), fragment/splice detection for images
Week 4	Pixel-Level Protection extended to documents (PDF rasterization pipeline), Master Admin Dashboard build-out, and codebase consolidation / merge-conflict resolution

PinithUB Platform & Employee Work Tracker — complete project work summary